

◇ Easy-Level Tasks

1. Create a Pod

- a. Task: Create a pod named `nginx-pod` running the `nginx:latest` image.
- b. Verify it's running with `kubectl get pods`.

2. Expose a Pod

- a. Task: Expose `nginx-pod` as a ClusterIP service on port 80.
- b. Confirm service endpoints with `kubectl get svc`.

3. ConfigMap Usage

- a. Task: Create a ConfigMap named `app-config` with a key `ENV=dev`.
- b. Mount it into a pod and verify with `kubectl exec`.

4. Secret Injection

- a. Task: Create a secret named `db-secret` with key `password=MyPass123`.
- b. Inject it into a pod as an environment variable.

5. Namespace Creation

- a. Task: Create a namespace called `exam-ns`.
- b. Run a pod inside that namespace.

◇ Mid-Level Tasks

6. Deployment Scaling

- a. Task: Create a deployment named `web-deploy` with 2 replicas of `nginx`.
- b. Scale it to 4 replicas and verify.

7. Rolling Update

- a. Task: Update the image of `web-deploy` from `nginx:1.19` to `nginx:1.20`.
- b. Ensure rollout status is successful.

8. Resource Limits

- a. Task: Create a pod with CPU request `100m` and memory limit `256Mi`.
- b. Verify with `kubectl describe pod`.

9. Node Selector

- a. Task: Schedule a pod only on nodes labeled `disk=ssd`.
- b. Add the label to a node and confirm scheduling.

10. Persistent Volume Claim

- Task: Create a PVC requesting `1Gi` storage.
- Attach it to a pod and verify the volume is mounted.

◆ Easy-Level Tasks (1–10)

1. Create a pod named `nginx-pod` using `nginx:latest`.
2. Expose `nginx-pod` as a ClusterIP service on port 80.
3. Create a ConfigMap `app-config` with key `ENV=dev` and mount it into a pod.
4. Create a Secret `db-secret` with key `password=MyPass123` and inject into a pod.
5. Create a namespace `exam-ns` and run a pod inside it.
6. Run a pod with a custom command (`echo Hello && sleep 3600`).
7. Create a pod with multiple containers (`nginx + busybox`).
8. Create a pod with a liveness probe checking `/healthz`.
9. Create a pod with a readiness probe on port 80.
10. Run a pod with an init container that prints “Init done”.

◆ Mid-Level Tasks (11–30)

11. Create a deployment `web-deploy` with 2 replicas of `nginx`.
12. Scale `web-deploy` to 4 replicas.
13. Perform a rolling update of `web-deploy` from `nginx:1.19` → `nginx:1.20`.
14. Roll back the deployment to the previous version.
15. Create a pod with CPU request `100m` and memory limit `256Mi`.
16. Schedule a pod only on nodes labeled `disk=ssd`.
17. Create a DaemonSet running `fluentd` on all nodes.
18. Create a Job that runs `echo "Hello Job"` once.
19. Create a CronJob that runs every minute printing “Hello Cron”.
20. Create a ServiceAccount `exam-sa` and assign it to a pod.
21. Create a PersistentVolume (PV) of 1Gi using `hostPath`.
22. Create a PersistentVolumeClaim (PVC) of 1Gi and attach to a pod.
23. Create a pod with a projected volume combining ConfigMap + Secret.
24. Create a pod with a downward API volume exposing pod name.
25. Create a pod with a node affinity requiring `region=us-east`.
26. Create a pod with a toleration for taint `key=dedicated:NoSchedule`.
27. Create a NetworkPolicy allowing traffic only from pods with label `role=frontend`.

28. Create a pod with a sidecar container writing logs to a shared volume.
29. Create a pod with ephemeral storage limit 500Mi.
30. Create a pod with a securityContext running as user 1000.

♦ Easy-Level Tasks (1–15)

1. Create a pod named busybox-pod that runs sleep 3600.
2. Run a pod with environment variable APP_MODE=production.
3. Create a pod with a volume mount using emptyDir.
4. Create a pod with a command override (echo "CKA Practice").
5. Create a pod with a restart policy Never.
6. Create a pod with a restart policy OnFailure.
7. Create a pod with a custom label tier=backend.
8. Create a pod with annotations owner=exam.
9. Create a pod with a lifecycle hook that prints "PreStop hook executed".
10. Create a pod with a liveness probe using exec command.
11. Create a pod with a readiness probe using HTTP GET /ready.
12. Create a pod with a startup probe checking /start.
13. Create a pod with a volume mount from a ConfigMap.
14. Create a pod with a volume mount from a Secret.
15. Create a pod with a projected volume combining ConfigMap + Secret.

♦ Mid-Level Tasks (16–30)

16. Create a ReplicaSet with 3 replicas of nginx.
17. Delete a pod and verify ReplicaSet recreates it.
18. Create a Deployment with rolling update strategy maxUnavailable=1.
19. Pause a Deployment rollout and then resume it.
20. Create a StatefulSet with 2 replicas using nginx.
21. Create a Headless Service for a StatefulSet.
22. Create a Job that runs date and exits.
23. Create a CronJob that runs every 5 minutes printing "Exam Task".
24. Create a pod with resource requests CPU=200m, memory=128Mi.

25. Create a pod with resource limits CPU=500m, memory=512Mi.
26. Create a pod with node affinity preferring zone=west.
27. Create a pod with pod affinity requiring pods with label app=frontend.
28. Create a pod with pod anti-affinity avoiding pods with label app=backend.
29. Create a pod tolerating taint dedicated=exam:NoSchedule.
30. Create a NetworkPolicy allowing ingress only from namespace exam-ns.